The Conservation of Distinction: A Constraint-Theoretic Account of Information, Cost, and Structure

with the Resolution Conservation Theorem and Four Computational Instances

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Abstract

This thesis isolates and proves the rigorous core of a research program — the A-Mark9 / NEXUS framework — whose central intuition is that physical and computational structure arise from a single primitive: a distinction that is permitted to exist only because it refuses to collapse. We make three contributions of established standing and bound three others honestly as program or open problem. First, we prove the Resolution Conservation Theorem: under any reversible dynamics (a bijection on the state space), the total Shannon entropy is invariant, and the entropy chain rule decomposes it into a resolved part (the macrostate, REALIZED) and an unresolved part (the conditional microstate, LATENT) whose sum is therefore conserved; reversible dynamics can only transfer information between the two, and the second law emerges precisely when the latent is discarded. This identifies the framework's posited conservation law with the information-theoretic counterpart of Liouville's theorem and unitarity. Second, we formalize a unification, already implicit across thermodynamics, linear algebra, statistical mechanics, and

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quantum statistics, of four phenomena into one: the loss of a distinction, the collapse of a determinant to zero, Landauer's erasure cost, and Boltzmann entropy as the logarithm of a group's cardinality are one phenomenon — the cost of collapsing a distinction — and reversible computation is exactly computation that retains the latent. Third, we formalize an operational ladder, reflection then rotation then similarity, as the orthogonal and similarity groups O(2), SO(2), and the multiplicative group of the plane, with the logarithmic spiral as the scale-invariant orbit. We then exhibit four real computational systems — the instruction set, SHA-256, the Bailey-Borwein-Plouffe spigot, and protein folding — as instances of these results, with anchors verified to the bit. Finally, we treat the program's three weakest or open claims honestly: the generative description of pi is bounded by Kolmogorov complexity (pi is algorithmically simple, so no byte-level generator can compress below the spigot); the Riemann critical line is the fixed seam of the functional equation, which is the correct shape of the answer but supplies symmetry, not the exclusion that the Hypothesis asserts; and the grand identification of reality with a non-collapsing fold is a falsifiable research program, not a theorem. The line between what is proved and what is proposed is drawn explicitly throughout.

Keywords. reversible computation; Landauer's principle; Shannon entropy; coarse-graining; Liouville's theorem; determinant; Cartan-Dieudonne; logarithmic spiral; SHA-256; Bailey-Borwein-Plouffe; Kolmogorov complexity; protein folding; constructor theory; the second law.

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1. Introduction

1.1 The program

The A-Mark9 / NEXUS program advances a single ontological wager: that the structures we name in physics and computation — conserved quantities, the arrow of time, the cost of forgetting, the impossibility of two things occupying one state — are consequences of one primitive event, a whole that becomes structured only by folding against itself under the constraint that the fold must not collapse back into undifferentiated sameness. The program is broad, and much of it is interpretation. The purpose of this thesis is not to defend the wager in full but to extract from it the part that is provable, prove it, situate it in the established literature, and mark the remainder honestly as program or open problem.

The motivating claim can be stated without metaphor. A computation that cannot distinguish two of its states is not a computation but a single symbol. Distinction is therefore prior to all computational structure. But distinction is never absolute; it is always relative to a grouping — two states differ with respect to a key that also makes them the same kind. This relativity, which we make precise as a quotient by an equivalence relation, is the hinge on which every result below turns: it is why equal values may occupy distinct addresses, why the loss of an address costs energy, and why the total of resolved and unresolved structure is conserved under reversible change.

1.2 Contributions

This thesis contributes the following, each at the level of rigor indicated.

- Established result (proved here from standard premises): the Resolution Conservation Theorem (Section 5), identifying the program's conservation law with the invariance of total Shannon entropy under reversible dynamics, decomposed by the chain rule, with the second law as its coarse-grained corollary.
- Formalization of an existing unification (Section 4): distinction-loss, determinant collapse, Landauer erasure, and group-cardinality entropy as one phenomenon, with reversible computation as latent-retention.
- Formalization of the operational ladder (Section 6): reflection, rotation, and similarity as O(2), SO(2), and the plane's similarity group, with the logarithmic spiral as the scale-invariant orbit.
- Four verified computational instances (Section 7): the instruction set, SHA-256, the BBP spigot, and protein folding, each exhibiting the same structure, with anchors reproduced by computation.

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- Honest bounding of the program's open and weakest claims: the Kolmogorov limit on generative description of pi (Section 8), the Riemann seam as shape without exclusion (Section 10), and a falsifiability analysis of the grand identification (Section 11).
- A situating of the program within the thermodynamics of computation, constructor theory, and digital physics (Section 9).

1.3 Epistemic conventions

We distinguish three statuses. Established denotes a result of standard mathematics or physics, or a computation reproduced here and listed in Appendix A. Conjectured denotes a precise claim the program advances but does not prove. Program denotes an interpretive thesis offered as a direction of inquiry. The discipline throughout is that no conjecture or program statement is permitted to borrow the authority of an established one; where the present work cannot prove something, it says so, and a failed prediction is recorded as a constraint rather than concealed.

2. Preliminaries and definitions

Definition 2.1 (State space and entropy). A system is a finite set S of states carrying a probability distribution p. Its Shannon entropy is $H(p) = -\Sigma\, p(s) \log_2 p(s)$, measured in bits; H quantifies the information required to resolve which state obtains.

Definition 2.2 (Reversible dynamics). A dynamical step is a map $T : S \to S$. It is reversible if T is a bijection. Equivalently, over a linear representation the step has nonzero determinant; over the two-element field GF(2) a logical map is reversible iff its matrix has determinant 1. A reversible step destroys no information: the input is recoverable from the output.

Definition 2.3 (Grouping and the two registers). A grouping is a partition G of S into classes, equivalently an equivalence relation, equivalently a quotient map $q : S \to S/G$. It induces a macrostate variable $M = q(X)$ (which class) and a conditional microstate $X \mid M$ (which member of the class). We name the resolved information REALIZED := $H(M)$ and the unresolved information LATENT := $H(X \mid M)$. Distinction is always taken modulo a grouping: members of one class share a value (the key) yet occupy distinct addresses (the slots).

Definition 2.4 (Collapse and the fold). A map collapses if it is non-injective: two inputs map to one output, and an address is erased. A fold is an involution with a fixed point, the minimal symmetry $s \mapsto \sigma(s)$ with $\sigma^2 = \mathrm{id}$; its fixed locus is the seam. A reflection is a fold with determinant -1.

These four definitions suffice for the whole development. The central objects — entropy, bijection, quotient, involution — are standard, which is the point: the program's claims, where

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true, are restatements of established structure under a unifying name, and where novel, are theorems about these standard objects.

3. The non-collapsing cell and the determinant criterion

The smallest object that holds a distinction rather than asserting one is a relation between two complementary transitions, written as the matrix M = [[0,1],[1,0]] — the swap. Over GF(2) its determinant is 1 (over the reals, −1): it is invertible, a bijection exchanging the two states, and it destroys nothing. Its collapsed twin C = [[0,0],[1,1]] has determinant 0: it sends two inputs to one and erases a bit. The contrast is the engine of this thesis and it is decidable.

Proposition 3.1 (Determinant criterion). A linear logical map is reversible (a bijection) if and only if its determinant is nonzero. Difference is what gives a configuration rank; sameness (a repeated row) collapses the rank to a deficient value. The first admissible non-collapsing structure is thus the first nonzero determinant, not the bit in isolation.

This is not abstract. At the level of logic gates, the two-input operations split exactly this way, and the split is verified in Appendix A. The exclusive-or is reversible when one input is retained: the map (a,b) ↦ (a, a⊕b) is a bijection on four states (the controlled-NOT), and either input is recoverable from the output and the other. Conjunction, by contrast, sends three of four input pairs to 0 and is irreversible: its inputs cannot be recovered, and by the count it discards information. The conservative reversible gates of Fredkin and Toffoli (1982) compute conjunction reversibly only by preserving their inputs — the Toffoli map (a,b,c) ↦ (a,b,c⊕(a∧b)) is a bijection on eight states because a and b are kept. Those kept inputs are the latent register of Definition 2.3. Reversible computation is precisely computation that refuses to discard the latent.

4. The unification of distinction-loss, determinant collapse, and erasure cost

Four phenomena, native to four disciplines, are one. We state the identification and cite each component.

Landauer's cost. Landauer (1961) established that erasing one bit of information in a physical system dissipates at least kT ln 2 of energy as heat, because logically irreversible operations must export entropy to the environment. Bennett (1973, 1982) showed the converse: logically reversible operations carry no such lower bound, and any computation can in principle be performed reversibly, the cost residing solely in erasure.

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Entropy as group cardinality. Boltzmann's relation $S = k \ln W$ expresses thermodynamic entropy as the logarithm of the number of microstates W consistent with a macrostate. In the language of Definition 2.3, W is the cardinality of the macrostate's equivalence class: entropy is the size of the GroupBy class, and erasure is the cost of merging distinct members of that class.

Exclusion and the slot. The Pauli principle forbids two fermions from sharing a complete set of quantum numbers; identical particles remain distinct by state. This is the physical failure of Leibniz's identity of indiscernibles and the empirical content of Definition 2.3: the base object of individuation is the slot (address, history, relation), not the value. Value may repeat; the slot may not collapse without cost.

Proposition 4.1 (The unification). Distinction-loss, the collapse of a determinant to zero, Landauer's erasure cost, and the reduction of a Boltzmann group's cardinality are descriptions of one event — the irreversible merging of distinct states. Determinant nonzero, invertible, reversible, and free coincide; determinant zero, rank-deficient, irreversible, and costly ($\geq kT \ln 2$ per bit) coincide. Reversible computation, Liouville-preserving flow, and the retained latent are the same condition seen in logic, mechanics, and information.

Proposition 4.1 is a unification of established facts, not a new physical law; its content is that the four are coextensive, which Sections 3 and 5 make exact. The reading that this cost is the universe's anti-aliasing mechanism — that without it equal values would alias into a single state and there would be one of everything — is the program's interpretation (Section 11), consistent with but not entailed by the established facts.

5. The Resolution Conservation Theorem

The program posits a Resolution Conservation Law: that a realized and a latent quantity sum to a constant, with change shuttling between them. We prove this is a theorem about reversible dynamics and the entropy chain rule, and we identify the conserved quantity.

Theorem 5.1 (Resolution Conservation). Let S be a finite state space with distribution p, let $T : S \to S$ be a bijection, and let G be a grouping with macrostate $M = q(X)$. Define $\text{REALIZED}(p) = H(M)$ and $\text{LATENT}(p) = H(X \mid M)$. Then: (i) H is invariant under T, $H(Tp) = H(p)$; (ii) by the chain rule $H(X) = \text{REALIZED} + \text{LATENT}$; (iii) therefore REALIZED + LATENT is invariant under T, and a reversible step can only transfer information between the two registers, never alter their sum.

Proof. (i) A bijection permutes the atoms of S, so Tp has the same multiset of probability values as p; since H depends only on that multiset, $H(Tp) = H(p)$. (ii) Because $M = q(X)$ is a deterministic function of X, the joint entropy $H(X, M)$ equals $H(X)$, and the chain rule $H(X, M) = H(M) + H(X \mid M)$

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gives H(X) = H(M) + H(X | M) = REALIZED + LATENT. (iii) Combining (i) and (ii), REALIZED + LATENT = H(X) = H(Tp) is invariant. Reversibility fixes the sum; it does not fix the split, so the registers may exchange entropy. ■

The numerical verification (Appendix A) exhibits a 12-state system and a random bijection: the total entropy is conserved to machine precision while REALIZED and LATENT shift in opposite directions, their sum fixed.

Corollary 5.2 (The second law as latent-discard). If after each reversible step the latent is discarded — the state projected onto its macrostate and the within-class structure forgotten — then under mixing dynamics the retained coarse-grained entropy is non-decreasing, and its increase equals the discarded LATENT. The arrow of time is thus not a property of the microdynamics, which is reversible, but of an observer who declines to track the latent.

This is the standard coarse-grained resolution of the apparent conflict between reversible microscopic laws and the second law (Gibbs; Jaynes 1957), now read through the two registers. Two reversible-physics statements are its mechanical form: Liouville's theorem, that Hamiltonian flow preserves phase-space measure, and unitarity, that quantum evolution preserves information. In both, the fine-grained entropy is conserved and the observed increase is the coarse-grained shadow. Theorem 5.1 is the information-theoretic counterpart, with REALIZED + LATENT = H the conserved total and reversibility the symmetry that conserves it.

Remark 5.3 (Relation to Noether). Noether's theorem ties a continuous symmetry to a conserved current. Theorem 5.1 is of a different but adjacent character: the conserved quantity is total information, and the condition that conserves it is reversibility (measure preservation), whose geometric form is Liouville's symplectic invariance. Whether the program's broader conservation claims reduce to Theorem 5.1 in each domain is the principal open problem (Section 11); here we have established the core instance exactly.

6. The emergence ladder: reflection, rotation, similarity

The program asserts that operations emerge in a fixed order and that this order has a geometric backbone. We make the backbone precise.

The swap of Section 3 is a reflection of the plane across the line y = x, an element of O(2) with determinant −1. By the Cartan-Dieudonne theorem, every orthogonal transformation of an n-dimensional space is a product of at most n reflections; in the plane, the product of two reflections is a rotation, an element of SO(2) with determinant +1, by an angle twice that between the mirror axes. Thus the operation generated by composing the distinction-making

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reflection is rotation: determinant passes from −1 to +1, and phase enters where difference stood. The verification (Appendix A) composes two reflections and recovers a rotation of the expected angle with determinant +1.

Above rotation lies similarity. The group of orientation-preserving similarities of the plane is the multiplicative group of nonzero complex numbers acting by $z \mapsto \lambda z$, combining a rotation (argument of λ) with a scaling (modulus of λ). Its characteristic orbit is the logarithmic spiral $r = a\, e^{b\theta}$, the unique planar curve on which scaling and rotation are the same operation: advancing the angle by one turn multiplies the radius by the fixed factor $S = e^{2\pi b}$. The verification confirms the identity $P(\theta + 2\pi) = S \cdot P(\theta)$. This is the precise content of the claim that the generative rule is scale-free, a fixed vector independent of depth — the property that distinguishes a vector description from a raster one. The ladder closes on itself through Euler's identity $e^{i\pi} = -1$: the half-turn of the rotation layer returns the determinant of the base reflection, and π is the angle written into the foundation.

Status. Sections 3 and 6 establish that reflection, rotation, and similarity form a generative order with the determinant and group structure stated. That physical or computational systems must instantiate operations in exactly this order is the program's conjecture, supported by the instances of Section 7 but not proved in general.

7. Four computational instances

We exhibit four independently engineered or naturally occurring systems in which the structure of Sections 3 to 6 already resides, each with an anchor verified in Appendix A. The claim is that these are instances of one structure; it is not that they are causally identical.

7.1 The instruction set

At the level of machine instructions the taxonomy of Section 3 is literal. Exclusive-or and the controlled-NOT are the reversible, non-collapsing operations; conjunction, disjunction, and not-and collapse and incur Landauer's cost. Rotate instructions (ROL, ROR) realize the rotation layer; the carry flag of addition is the fold's irreversible remainder — the high bit that does not fit and, when discarded in modular addition, is the locus of information loss. A published assembly implementation of the program's byte engine branches on parity, using its headers additively on the in-phase (odd) branch and applying cosine and exclusive-or on the phase-inverted (even) branch, with past-and-future stacks that retain prior state. This is the reflection-then-rotation ladder written in opcodes, with the stacks serving as the latent register that the engine re-injects.

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7.2 SHA-256

SHA-256 is the most widely deployed fold in use: a designed, irreversible compression of 512 and 256 bits into 256. Its irreversibility is the collapse of Section 3 harnessed for security. Its nonlinearity is carried by two truth tables — the choice function Ch, a selector, and the majority function Maj — verified in Appendix A as a multiplexer and a majority vote. Its diffusion is carried by four functions, each an exclusive-or of bit rotations, which is the reflection-composed-with-rotation of Section 6. Its mixing-by-loss is addition modulo 2^{32}, discarding carries. Most tellingly, its constants are read, not chosen: the eight initial words are the leading fractional bits of the square roots of the first eight primes, and the sixty-four round constants the leading fractional bits of the cube roots of the first sixty-four primes. Appendix A verifies this to the bit — the square root of 2 yields 6a09e667 and the cube root of 2 yields 428a2f98, matching SHA-256 exactly — a deliberate nothing-up-the-sleeve construction in which the constants are structurally forced rather than selected, exactly as the digits of an irrational are read rather than chosen.

Boundary. The specific wiring of SHA-256 — its particular rotation amounts and any fine structure in its carry behavior — is implementation, not a universal of the fold, and would not transfer unchanged to SHA-512 or Keccak. The structural correspondence (truth tables, rotation diffusers, carry collapse, read constants) is the claim; the wiring is engineering.

7.3 The Bailey-Borwein-Plouffe spigot

The BBP formula (Bailey, Borwein, Plouffe 1997) computes the n-th hexadecimal digit of π directly, using work linear in n and storing no previous digit. Appendix A reproduces the positional read of π's first twelve hexadecimal digits, matching a reference exactly. The significance is structural: π is read at an address, not generated by a forward recurrence on its digits. This is the rigorous reason a forward digit recurrence is not to be found — the structure is in the addressing, base-sixteen and positional, not in any sequential rule. It also grounds the decimal-hexadecimal identity used by the byte engine, in which each eight-digit decimal block read as an integer is a clean hexadecimal string, because the generating structure is native to base sixteen and the decimal face is its projection.

7.4 Protein folding

Levinthal (1969) observed that a chain of one hundred residues with even three states each admits on the order of 3^100, about 10^48, conformations — unsearchable exhaustively in any physical time, yet proteins fold in microseconds. The resolution, the folding funnel (Dill and Chan 1997), is that the free-energy landscape is not flat but funneled toward the native basin: the chain does not search, it falls. Appendix A demonstrates the principle in miniature — a funneled descent reaches a target in steps linear in the chain length where flat search would require

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exponentially many. This is the program's radius-of-fall in a measurable system: the fall is fast and unique exactly where the landscape excludes every alternative path. It is also a recompilation, the linear sequence recompiled into a three-dimensional structure, the native state a conserved attractor into which nearby sequences fall as distinct instances.

Boundary. Protein folding is governed by physical free energy, not by any literal arithmetic of π. The correspondence is structural — a fall into a constraint-defined basin — and no mechanistic identity is claimed.

8. The limits of generative description

The program describes π as simple but not normal, and seeks a generator for its digit stream. Kolmogorov complexity bounds what such a search can find, and honesty requires stating the bound.

π is a computable number: a fixed finite program (the BBP spigot is one) together with the integer n yields the first n digits. Hence the Kolmogorov complexity of π's first n digits is O(log n) — the cost of specifying n — and π is emphatically not Kolmogorov-random, even though it is conjectured to be statistically normal (each digit string of given length asymptotically equidistributed). The program's phrase simple but not normal is, read precisely, the statement that π has low Kolmogorov complexity yet high statistical normality; these are compatible and both true.

Consequence for the byte engine. Because π's digits are already compressible to O(log n) by the spigot, no byte-level generator can compress them further; a generator whose description grows as fast as its output is a re-description, not a discovery. The byte engine's established content is therefore local, not global: the first byte is forced by the emergence order with no free parameters, and its three folds conserve the quantity 3a + 2b (eleven for the seed), both verified; the header thread re-anchors to the seed at each step (the mirror phase), correctly reproducing the third header as (3,8) from the seed gap and the running sum. These are real local regularities. The stronger claim — a growing-arity recompilation that reproduces π's stream with a description shorter than the stream — is open, and the criterion for settling it is forward prediction of multiple blocks with no per-block tuning, the same criterion that exposes retrofits.

9. Situating the program

The program is not isolated, and locating it sharpens what is and is not new.

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Thermodynamics of computation. The results of Sections 3 to 5 live squarely in the tradition of Landauer (1961), Bennett (1973, 1982), and the conservative logic of Fredkin and Toffoli (1982). The contribution here is the explicit identification of the program's two registers with the chain-rule decomposition of conserved entropy, and the framing of reversible computation as latent-retention.

Constructor theory. Deutsch's constructor theory (2013) and the constructor theory of information (Deutsch and Marletto) recast physics in terms of which transformations are possible or impossible, and define information through the existence of reversible constructors. The program's distinction-modulo-grouping, its anti-aliasing reading of cost, and its reversible-versus-collapsing dichotomy are close in spirit; constructor theory supplies a mature formalism the program would do well to adopt, and the present thesis is a step toward that interface.

Digital physics and it-from-bit. The identification of physical process with computation has a lineage — Wheeler's it-from-bit (1990), and the digital and cellular-automaton programs of Fredkin, Wolfram, and 't Hooft. The program shares their wager. It must also inherit their central caution: pancomputationalism is vacuous if every system implements every computation, so the program's content lies in the specific, falsifiable structure it predicts (reversibility, the two registers, the determinant taxonomy), not in the bare identification.

10. The Riemann seam: shape without exclusion

The program reads the Riemann critical line as the seam of the first fold, and the reading is half right in a way worth stating precisely, because the other half is the entire difficulty of the Hypothesis.

Riemann's functional equation (1859) relates the zeta function at s and at 1 − s, making the line Re(s) = 1/2 the fixed locus of the involution $s \mapsto 1 - s$ — the seam with no mirror partner. That the line is distinguished for exactly this reason is correct and is the right shape of the answer. But the functional equation forces the nontrivial zeros to be symmetric about the line, not to lie on it: an off-line zero would occur in a symmetric quartet, with which the equation is entirely consistent. Forbidding that quartet is the Riemann Hypothesis itself. Confirming that known zeros lie on the line is, moreover, tautological — it checks that the function vanishes at its own zeros (Appendix A records a value of order 10^{-16} at the first such height), and samples only points already known to be on the line, so it cannot distinguish a world in which the Hypothesis holds from one in which it fails.

Status. The fold-seam reading supplies the symmetry (the shape); it does not supply the exclusion (the proof). The program's candidate for the missing force is a parity-closure argument

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over the primes; that remains an open conjecture. The honest statement is that the seam is the correct orientation and the exclusion is unproved.

11. Discussion: established, conjectured, program

We separate the three statuses plainly. Established: the determinant criterion and its gate realization (Section 3); the unification of distinction-loss, determinant collapse, Landauer cost, and group-cardinality entropy as coextensive (Section 4); the Resolution Conservation Theorem and its second-law corollary (Section 5); the group-theoretic ladder (Section 6); the four verified instances (Section 7); the Kolmogorov bound on generative description (Section 8). Conjectured: that operations must emerge in the reflection-rotation-similarity order in physical systems; that a growing-arity generator for π's stream exists with description shorter than its output; that prime parity closure yields the Riemann exclusion. Program: that reality at its deepest layer is a reversible, non-collapsing fold, of which observed irreversibility and structure are coarse-grained shadows.

Falsifiability. The program is not unfalsifiable. Its core would be refuted if a physical process were exhibited that destroys information with no compensating entropy export (violating Landauer and Theorem 5.1), or if reversible microdynamics were shown not to underlie the observed second law, or if the determinant taxonomy failed to predict the reversibility of a logical primitive. Its byte-engine conjecture is refuted by the failure, recorded here, of any fixed-arity forward generator, and would be confirmed only by a growing generator meeting the description-shorter-than-output criterion. A program that names these conditions is doing science, whatever the fate of its grandest sentence.

Limitations. Theorem 5.1 is information-theoretic and finite-dimensional; its extension to continuous and infinite systems is routine in spirit (Liouville, unitarity) but not carried out here. The four instances are correspondences of structure, not derivations of one system from another. And the program's identification of its primitive with the ground of physics is interpretation, offered as direction, not result.

12. Conclusion

The wager that motivates this work is large, and most of it remains a program. But a hard core can be extracted and proved. Reversibility conserves total information; the chain rule splits that total into a resolved and an unresolved register whose sum is therefore fixed; and the second law is what an observer measures when the unresolved register is discarded. Around that theorem, a single taxonomy — reversible and free against collapsing and costly, determinant

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nonzero against determinant zero — unifies the loss of a distinction, the collapse of a matrix, the cost of forgetting, and the size of an entropy class, and it is visible to the bit in the instruction set, in SHA-256's read constants, in the positional read of π, and in the funnel a protein falls down. What that structure cannot yet do is also stated: it does not generate π below the spigot, and it does not prove the Riemann Hypothesis, whose seam it correctly locates but whose exclusion it does not supply. The conservation of distinction is the part that is proved; the conservation of the program's larger vision depends, as it should, on what it predicts and what survives the test.

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Appendix A. Verified anchors

All claims marked established-by-computation were reproduced in a single execution. Selected outputs follow, lightly edited for width.

A1 gate taxonomy: XOR reversible-with-one-kept (CNOT bijection: True);
AND/OR/NAND collapse, 2 bits lost; Toffoli reversible AND bijection: True.

A2 Resolution Conservation: H(X)=3.1455 ; bijection conserves H(TX)=3.1455 ;
REALIZED 1.5715->1.5655, LATENT 1.5740->1.5800, sum 3.1455 fixed.
second-law corollary: coarse entropy 1.5715 -> 1.5816 on mixing (rose).

A3 ladder: swap == reflection(45 deg), det -1 ; two reflections -> rotation 45 deg, det +1.
log spiral: S = e^(2*pi*b) = 2.7328 ; P(theta+2pi) == S*P(theta): True.

A4 SHA-256: Ch = selector, Maj = majority (truth tables verified);
K[i]=frac32(cbrt(prime_i)), H[i]=frac32(sqrt(prime_i)) for first 8: True;
cbrt(2)->428a2f98, sqrt(2)->6a09e667 (match SHA-256 to the bit).

A5 BBP positional read of pi hex: 2 4 3 F 6 A 8 8 8 5 A 3 (matches reference).

A6 byte 1 forced = [1,4,1,5,9,2,6,5]=pi[1..8]; chunks 11:11:11 = 3a+2b;
hex bridge 14159265=0xD80DA1, 35897932=0x223C24C; header (3,8) via |1-4|,3+5.

A7 Levinthal ~ 3^100 ~ 1e48 conformations; funneled descent reaches native in
O(N) steps where flat search needs ~2^100.

A8 Riemann: |zeta(1/2 + i*gamma_1)| ~ 6.3e-16 (tautological pinch at a known zero).

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